PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

IDEAL MVP LABOR FORCE, MACHINERY & 12-MONTH BUFFERED TIMELINE

Sovereign Bio-Solar-Thermal Campus | JET EOI Operational Execution Specification

Doc Ref: 24_labor_machinery_timeline.html | Status: JET EXECUTION BASELINE (20,000M² FENCE INFUSED) | Governance: Scapush Precision | 12-Month Gantt Master

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1. Human Capital & Machinery Execution Strategy

LISTEN SECTION

This document quantifies the exact labor force man-hours, skilled trade allocation, heavy/specialized machinery plant-hours, and a 12-month buffered execution timeline for the **R3.73M Scaled MVP Scope (Infused with 20,000m² Perimeter Fence)**, featuring 4,800 Broilers, 160 Swine, 70m³ Biodigester, 30 kWp Solar PV, 15 kW Biogas CHP, 60 kWh LFP BESS, Accessibility Hub, and SCADA Control Center.

PRECISION RESOURCE ALLOCATION & 12-MONTH BUFFER MANDATE

Civil works commence with the 20,000m² site perimeter fencing (2.2m height) and earthworks to establish immediate sovereign biosecurity control. Construction emphasizes localized labor trade crews and targeted machinery deployments, incorporating a 2.5-month weather and supply-chain contingency buffer for on-time JET EOI commissioning.

2. Labor Force Breakdown & Trade Man-Hours Register

LISTEN SECTION

Labor force requirements are calculated based on standard South African civil and MEP productivity rates adjusted for high-precision off-grid nexus installation:

Build Phase / Trade Category	Skill Level & Role Title	Peak Personnel Count	Total Phase Man-Hours	Primary Work Packages & Deliverables
Phase 01: Fence, Civil & Water	Site Supervisor, Fencing Crew, Rig Operators, Laborers	10 Personnel	**920 Man-Hours**	20,000m² perimeter security fence (2.2m height), borehole drilling oversight, trenching, 6m tank tower erection, concrete pads.
Phase 02: Solar & Microgrid	Certified Wireman, PV Installers, Riggers	6 Personnel	**560 Man-Hours**	30kWp PV mounting frames, panel wiring, 60kWh BESS positioning, 30kVA hybrid inverter AC/DC integration.
Phase 03: Barns 1 & 2	Steel Erectors, Bricklayers, Plumbers	12 Personnel	**1,840 Man-Hours**	Barn 1 (384m²) & Barn 2 (384m²) structures, 2% sloped floors, gravity flumes, drinker/feeder lines, biosecurity station.
Phase 04: Digester & CHP	Masonry Specialists, Gas Fitters, Mechanics	8 Personnel	**1,120 Man-Hours**	70m³ fixed-dome masonry dome, gas scrubbing manifold, 15kW CHP set placement, hydronic heat

Build Phase / Trade Category	Skill Level & Role Title	Peak Personnel Count	Total Phase Man-Hours	Primary Work Packages & Deliverables
				exchanger piping.
Phase 05: Nursery & SCADA	Instrumentation Tech, Electrician, HVAC	5 Personnel	**480 Man-Hours**	Hydronic floor loop manifold, Zone F nursery pod insulation, SCADA PLC desk wiring, 120 sensor calibrations.
Phase 06: Access & Academy	Paving Crew, Carpentry, General Labor	6 Personnel	**400 Man-Hours**	Zone M step-free paving, 4x wheelchair bays, 30kW DC EV charger mount, 12m container classroom fit-out.
Project Commissioning	Lead Systems Engineer, SCADA Specialist	3 Personnel	**320 Man-Hours**	End-to-end nexus integration, black-start testing, biosecurity wet runs, Academy operational handoff.
	TOTAL REQUIRED MVP LABOR MAN-HOURS:		**5,640 Man-Hours (Avg. 8–12 Workers on site)**	

3. Machinery, Heavy Plant & Specialized Equipment Hours Register

LISTEN SECTION

Machinery hours are scheduled to maximize machine efficiency, avoiding idle standby costs on site:

Machinery / Equipment Type	Capacity / Machine Specification	Operating Hours	Primary Site Tasks & Execution Purpose
Hydraulic Rotary Drilling Rig	Heavy Truck-Mounted Rig (100m depth capability)	**32 Machine-Hours**	Drilling Borehole-01 (165mm uPVC casing) to water strike and gravel packing.
20-Ton Crawler Excavator	Bucket capacity 0.9 m^3, Breaker attachment	**64 Machine-Hours**	Earthworks site leveling, 70m^3 digester underground bowl excavation, slurry trenching.
TLB (Tractor Backhoe Loader)	4x4 Heavy Utility TLB	**120 Machine-Hours**	Foundation trenching for Barns 1 & 2, water pipe/cable trench backfilling, fence post holes.
10-Ton Smooth Drum Compactor	Vibratory Soil Roller Compactor	**32 Machine-Hours**	Sub-base compaction for Barn concrete slabs, Zone M paving, and digester base.
Mobile All-Terrain Crane	25-Ton Mobile Hydraulic Crane	**24 Machine-Hours**	Lifting 6m steel water tank tower, 12m container classroom, BESS enclosure, CHP set.
Ready-Mix Concrete Trucks	6 m^3 Drum Transit Mixers (18 Trips Total)	**48 Plant-Hours**	Pouring Barn slabs (768m^2 total), biosecurity station pad, digester base, tank footings.
Plate Compactors & Mixers	On-site 350L Concrete Mixers & Rammer Compactors	**210 Equipment-Hours**	In-situ masonry mortar mixing for digester, localized infill compaction, pathway edge beams.
	TOTAL DEDICATED MACHINERY & PLANT TIME:	**530 Machine/Plant Hours Across Build Lifecycle**	

4. 12-Month Buffered Project Master Timeline (Chronological Gantt)

LISTEN SECTION

A 12-month timeline structured into distinct milestones, incorporating a 10-week contingency buffer for wet-weather delays and import lead times:

12-Month Chronological Execution Schedule

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M01: 20k m² Perimeter Fence, Borehole-01 & Site Civil Establishment
                                    ↓
M02--M03: Water Tower Reticulation, 70m³ Digester Masonry Bowl & Slurry Flumes
                                    ↓
M04--M05: Barn 1 (Poultry 4.8k) & Barn 2 (Swine 160) Superstructures
                                    ↓
M06--M07: 30kWp Solar PV, 60kWh BESS, 15kW CHP & Power Microgrid
                                    ↓
M08--M09: Zone F Nursery Hydronics, SCADA 120-Tag Logic & Zone M Access Hub
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M10--M12: [ BUFFERED PERIOD ] Wet Testing, Biosecurity Stocking & JET Handoff
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Month	Primary Work Package	Labor & Machinery Intensity	Milestone Output & Buffer Safeguard
Month 01	20k m^2 Perimeter Fence & Borehole Drilling	1 Rig, 1 TLB, 10 Workers (400h)	2.2m high security fence erected around 20,000m^2 leasehold; Borehole-01 struck & cased.
Month 02	Water Tower Reticulation & Civil Earthworks	1 TLB, 1 Compactor, 8 Workers (360h)	Elevated 20kL tank tower erected; site pad leveled; primary utility trenching complete.
Month 03	70m^3 Digester Dig & Masonry Core Build	1 Excavator, Bricklayer Crew (520h)	Digester bowl excavated and dome masonry constructed with gas-tight seal layer.
Month 04	Barns 1 & 2 Foundations & Slurry Flumes	Ready-Mix Trucks, 12 Workers (640h)	Barn 1 (384m^2) and Barn 2 (384m^2) concrete slabs and 2% sloped flumes completed.
Month 05	Barn Superstructures & Biosecurity Station	Steel Erectors, Plumbers (680h)	Roofing, wall cladding, drinker/feeder lines, and shower-in/out station installed.
Month 06	30 kWp Solar PV & Mounting Array	PV Electrical Crew, Riggers (360h)	Ground frames and rooftop arrays mounted; DC combiner boxes wired.
Month 07	BESS Enclosure & 15 kW CHP Set Delivery	Mobile Crane, Wiremen (400h)	60kWh LFP battery container placed; 15kW CHP engine aligned with exhaust heat loops.
Month 08	Microgrid Switchgear & Hydronic Piping	Certified Electricians, Plumbers (440h)	30kVA hybrid inverter energized; nursery hydronic underfloor loops connected.
Month 09	SCADA Control Desk & Sensor Wiring	Instrumentation Engineer (320h)	120 telemetry sensors online; PLC programmed; HMI dual monitors rendering live maps.
Month 10	Zone M Accessibility Hub & Academy Lab	Paving Crew, Carpenters (360h)	Step-free pathways paved; 30kW DC EV charger mounted; Academy container ready.
Month 11	**[BUFFER 1] Integrated Wet-Commissioning**	Lead Engineers, Operators (200h)	Full bio-water-power system wet runs, black start tests, and biodigester seeding.
Month 12	**[BUFFER 2] Biosecurity Stocking & Handover**	Academy Staff, Operators (200h)	4,800 chicks & 160 swine stocked; JET EOI audit verification & Academy launch.

5. Operational Lock-In Register & JET EOI Execution Declaration

LISTEN SECTION

Ref Code	Locked Operational Parameter	Governance & EOI Execution Basis
EXEC-001-F	20,000m^2 Perimeter Fencing Execution	Immediate Month 1 boundary lockdown with 2.2m high heavy-duty security fence.
EXEC-002	5,640 Total Labor Man-Hours	Managed across 6 distinct specialized trade crews over 12 months.
EXEC-003	530 Machinery / Plant Hours	Optimized machine deployment eliminating idle equipment rental costs.

Ref Code	Locked Operational Parameter	Governance & EOI Execution Basis
EXEC-004	12-Month Schedule (2.5m Buffer)	Robust timeline built to absorb Highveld rainy season delays and import lead times.

6. Approval & Authorization

LISTEN SECTION

DOCUMENT STATUS

STATUS: APPROVED BASELINE (REV 2.0) — INFUSED WITH 20,000M² PERIMETER FENCE & LOCKED FOR JET EOI EXECUTION

PROJECT STEWARD & TECHNICAL LEAD



Select File | Lock Signature

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Consortium Leadership

Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
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Execution Entities & Standard

Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint

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